

Enhanced Gibbs Sampling for Transcription Factor Binding Site Discovery in *M. tuberculosis*

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Goal

The goal of this project is to improve GibbsSampler by reducing convergence to local optima and comparing modified versions on real tuberculosis DNA data to evaluate which methods improve motif-finding performance.

Background

- Motif finding identifies short DNA patterns such as transcription factor binding.
- GibbsSampler is a randomized motif-finding algorithm that iteratively updates motifs using a probability profile.
- We evaluated simulated annealing, random restarts, greedy warm starts, and entropy selection algorithms.
- Applied methods to tuberculosis genomic data.

ELSI

Ethical, legal, and social implications issues were not directly applicable to our project as we used publicly available tuberculosis genomic data from established databases rather than sensitive human data.

Algorithm

Baseline Gibbs

A Gibbs sampler was implemented following the Rosalind problem set formulation [1]. At each iteration a random sequence is excluded, a profile is built from the remaining motifs, and a new k-mer is sampled proportional to that profile. Every proposed k-mer is unconditionally accepted, with the best motif set tracked separately across all iterations.

```
For j = 1 to N:  
  i ← random sequence  
  profile ← ProfileBuilder( motifs \ motif_i)  
  motif_i ← profileRandomKmer(sequence_i, profile)  
  If Score(motifs) < Score(best)  
    best ← motifs
```

Simulated Annealing

A random k-mer injection (20% of iterations) forces exploration beyond the self-reinforcing profile. Metropolis Hastings acceptance filters both random and profile-guided proposals, blocking destructive jumps from corrupting good solutions already found.

```
delta = Score(candidate) - Score(current)  
if delta < 0 or rand() < exp(-delta/T):  
  accept motif_i  
T ← T x exp(log(0.01)/N)
```

Greedy Warm Start

Instead of random initialization, we run an exhaustive greedy motif search. This warm start gives SA a better initial solution. Temperature is lowered since less exploration is needed from a good starting point. SA is kept due to its success in improving ground truth match,

```
for each kmer in seq 1:  
  greedily build motifs  
keep best scoring motif set  
Run Simulated Annealing gibbs sampler
```

Entropy Selection

Replaces the Hamming distance score with Shannon entropy, which captures partial conservation across all four bases per column rather than just counting mismatches from the majority nucleotide.

```
# Score Function  
for each column:  
  for each base b:  
    p = count_b / t  
  H -= p * log2(p)
```

Experiment

Data:

- 14 DosR binding sites (CollecTF [3], H37Rv [2])
- 14 × 520bp (extracted) input sequences, 20bp ground truth sites
- k=20, N=2000, 30 restarts

Metrics:

- Score: counts mismatches from consensus per column (Hamming distance) → lower is better
- Entropy score: per-column Shannon entropy $H = -\sum p \cdot \log_2(p)$, lower is better
- Ground truth: pairwise match matrix with optimal one-to-one assignment to maximize total match percentage

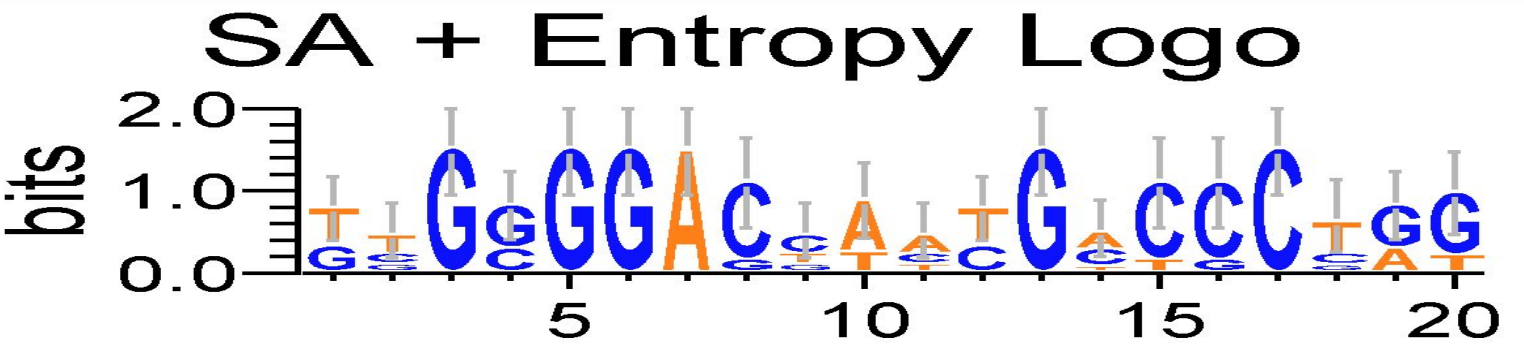
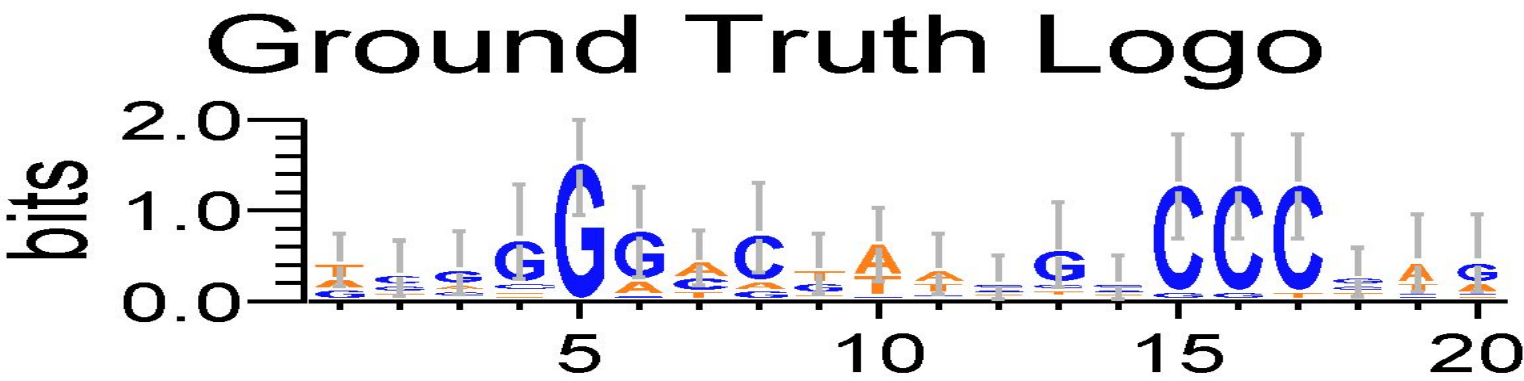
Results:

- Ground truth: Hamming 81, entropy 23.65, GT match 100%, upper bound for biological accuracy
- SA Gibbs: Hamming 71.20, GT match 61.4%, more conserved than ground truth but not biologically accurate
- Greedy + SA: best Hamming (63) but lowest GT match (33.9%), greedy initialization biases search toward early local optima
- Entropy and Augmented: entropy 15.15/15.51 vs ground truth 23.65, over-conserved. They peak at 63.2% GT match but some runs score as low as 30%, as entropy cannot distinguish a highly conserved wrong motif set from the correct one
- No model consistently recovers known binding sites across 30 restarts, scoring functions cannot distinguish true motif from other conserved sequences in 520bp windows

Algorithm	Avg Score	Avg Entropy	GT Match %
Ground Truth	81	23.65	100
Baseline	76.10	N/A	23.1
SA Gibbs	71.20	N/A	61.4
Greedy + SA	63	N/A	33.9
Entropy	N/A	15.15	63.2
Augmented (SA + Entropy)	N/A	15.51	63.2

Conclusion:

- SA + Entropy Model identifies the GGG/CCC core (confirmed by sequence logos) but over-conserve flanking positions, converging on a more specific k-mer than the true binding sites
- Conservation-based objectives cannot distinguish the true motif from other conserved sequences in the 520bp window, a wrong but highly conserved k-mer scores equally well as the correct one
- GT match peaking at 63.2% suggests unsupervised conservation alone is insufficient; a position-specific scoring matrix anchored to known sites would likely improve biological accuracy



References

- [1] Compeau, P.E.C. and Pevzner, P.A., Bioinformatics Algorithms: An Active Learning Approach, Vol. 1, Active Learning Publishers.
- [2] NCBI Nucleotide Database. Mycobacterium tuberculosis genome (NC_018143.1). National Center for Biotechnology Information. https://www.ncbi.nlm.nih.gov/nuccore/NC_018143.1
- [3] CollecTF Database. Transcription factor binding site data (EXPSITE_000013c0). http://www.collectf.org/EXPSITE_000013c0